Speaker

DO | PICO | OUTPUT | SOUND

What does it do?

A small passive speaker turns a changing electrical signal into sound.

Current through its coil moves a cone backwards and forwards, pushing the air.

The 220 ohm resistor shown here protects the GPIO; it also keeps this simple circuit fairly quiet.

Uses



Door chime

DETECT Button and chime setting → **DECIDE** Which chime should play? → **DO** Play the notes



Tiny fanfare machine

DETECT Count achievement-button presses → **DECIDE** Has the target been reached? → **DO** Play triumphant notes



Cone
Pushes the air

Coil and magnet
Create movement

Terminals
Connect through 220 ohms

Raspberry Pi Foundation

Code

Raspberry Pi Pico

```
from picozero import Speaker

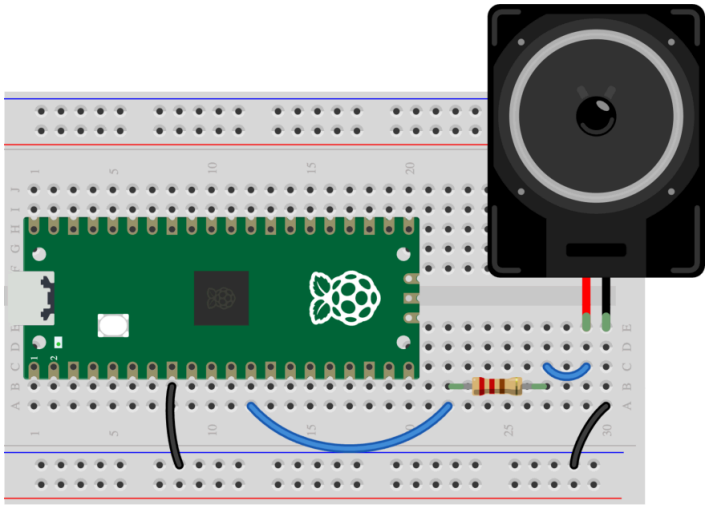
speaker = Speaker(9)

speaker.play("E4", duration=1)
```

For a louder speaker, use a transistor or amplifier rather than removing the protective resistor.

Wiring

PICO



GP9 (pin 12) → 220 ohms → speaker red (+) lead; black lead → **GND** (pin 8) via the − rail